Chapter 1 — Measurements

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Significant figures in 0.0025: (a) 3 (b) 5 (c) 2 (d) 4

2. 43.75 to 3 s.f. (even rule): (a) 43.8 (b) 43.75 (c) 43.7 (d) 44.0

3. 2.8723 × 1.6 to correct s.f.: (a) 4.60 (b) 4.6 (c) 4.5957 (d) 4.596

4. In addition, keep least: (a) decimal places (b) units (c) prefixes (d) significant figures

5. Percentage uncertainty is: (a) $\Delta x \times 100$ (b) $\Delta x + x$ (c) $x/\Delta x \times 100$ (d) $\Delta x/x \times 100$

6. For $A = \pi r^2$, % uncertainty in A: (a) ½ × % in r (b) 4 × % in r (c) 2 × % in r (d) % in r

7. Dimensions of power: (a) $[ML^{-1}T^{-2}]$ (b) $[ML^2T^{-3}]$ (c) $[ML^2T^{-2}]$ (d) $[MLT^{-3}]$

8. Precision depends on: (a) true value (b) least count (c) unit name (d) percentage only

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why does every measurement contain some uncertainty?

2. How is the uncertainty of repeated readings estimated?

3. Why are leading zeros not counted as significant figures?

4. How does scientific notation remove ambiguity in significant figures?

5. Why can a precise measurement still be inaccurate?

6. Why are absolute uncertainties added in subtraction?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain absolute, fractional and percentage uncertainty with an example. **(4)**

(b) Derive the rules for combining uncertainties in sums, products and powers. **(4)**

Q.4 (a) Differentiate between precision and accuracy and relate them to uncertainty. **(4)**

(b) Explain the uses and limitations of dimensional analysis. **(4)**

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